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NPS/003 /008 – Technical Specification for 66kv and 132kv Circuit Breakers

1. Purpose

This document is the technical specification for 66kV and 132kV circuit breakers to be utilised on the NEDL and YEDL distribution networks.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/003/008	Version 1	Technical Specification for 66kv and 132kv Circuit Breakers

2. Scope

This specification details the technical requirements for 66kV and 132kV open terminal circuit breakers to be utilised on the NEDL and YEDL networks. It will be necessary to consider and include any project specific requirements as detailed in Appendix 3, Addendum To Supplier Requirements.

3. Technical Requirements

All circuit breakers shall comply fully with the requirements of the latest version of Energy Network Association Technical Specification (ENATS) 41-37 and the latest versions of standards and regulations to which ENATS 41-37 refers.

The circuit breakers will preferably have been assessed by the ENA Switchgear Assessment Panel (SAP) and have been awarded an ENA Notice of Conformity.

Appendix 1 of this document details the requirements selected within ENATS 41-37, or where the requirements of ENATS 41-37 have been supplemented. Note particularly the specified fault-break duties and X/R ratios.

Appendix 2 shall be completed by the manufacturer to confirm the technical details of the circuit breaker.

Appendix 3 contains an Addendum to Supplier Requirements which lists project specific requirements.

4. References

4.1 External Documentation

Reference	Title
ENATS 41-37	Switchgear for use on 66kV to 132kV Distribution Systems
BSI 381C 1996	Specification for colours for identification, coding and special purposes.

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4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K.Midgley	Standards Engineer	K.Midgley	23/05/06

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G.Bolden	Purchasing Manager	G.Bolden	23/05/06

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D.Gazda	Standards Team Co-ordinator	D.Gazda	23/05/06

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	07/09/06

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	23/05/06

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	25/05/06

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Appendix-1: Technical Specification

66kV Circuit Breaker

	ENATS 41-37 Reference	UNIT	
Particular of System			
Voltage		kV	66
Frequency		Hz	50
Number of Phases			3
Neutral Earthing			Solid
Circuit Breaker Characteristics			
Number of poles			3
Class – indoor/outdoor	1.2		Outdoor
Rated voltage	4.1	kV	72.5
Rated Insulation level. Lightning impulse level	4.2	kV	325 375 (across isolating gap)
Power frequency withstand			140 160 (across isolator gap)
Rated Frequency	4.3	Hz	50
Rated Normal Current	4.4	A	Feeder circuit breakers: 1250A Bus section, bus coupler, and transformer circuit breakers: 2000A
Rated Short-time withstand current	4.5	kA	31.5
Rated peak withstand current	4.6	kA	78.75
Rated duration of short circuit	4.7	S	3
Rated supply voltage of opening and closing devices and auxiliary and control circuits. Closing and tripping Indication Control	4.8	V	110 110 110
Rated supply frequency of closing and auxiliary circuits.	4.9	Hz	DC
Rated short-circuit breaking current. Equal to rated short-time withstand current.	(Part 3) 4.101	kA	31.5 @ X/R of 14
Rated short-circuit making current	(Part 3) 4.103	kA	Manufacturer to state
Rated operating sequence.	(Part 3) 4.104		0 – 0.3 s – CO – 3min – CO

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Characteristics for short line faults	(Part 3) 4.105	kA	Manufacturer to state
Out of phase making and breaking.	(Part 3) 4.106	kA	Manufacturer to state
Rated capacitive switching currents : Rated cable-charging current.	(Part 3) 4.107	A	Manufacturer to state but should be Class 2
Rated small inductive breaking current.	(Part 3) 4.108	A	Manufacturer to state
Rated DC time constant.	(Part 3) 4.109	Msec	Manufacturer to state (see specified fault breaking current)
Classification of mechanical operations M1 or M2	(Part 3) 4.110		M1
Forced opening: Yes/No	(Part3)5.4.103		Manufacturer to state
Anti pumping: Yes/No	(Part3)5.5.101		Manufacturer to state
Mechanism type			Manufacturer to state
Independent drive mechanism:	(Part3)5.5.103		Manufacturer to state
Local manual tripping: Yes/No	(Part3)5.8.201		Manufacturer to state
Local manual closing: Is this facility provided? Yes/No	(Part3)5.8.202		Manufacturer to state
External bushing creepage distance	5.14		As per 41-37 Pt 1 (25mm per kV)
External bushing pollution performance	5.14.101		As per 41-37 Pt 1 (Class III)
Personal safety clearance	5.107		As per 41-37 Pt 1 (i.e. additional 300mm)
Air insulated bushings	5.108		As per 41-37 Pt 1
Noise	(Part 3) 5.203	dB	Manufacturer to state
Provision for key interlock			Provision for 1 key
CT secondary terminal			Spring loaded or "cage clamp" design. Spring washers on stud type
Bus zone CT secondary terminals.			Shorting/disconnecting facility.
Scada terminals			Knife terminal equipped with test 2.3mm sockets

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132kV Circuit Breaker

	ENATS41-37 Reference	UNIT	
Particular of System			
Voltage		kV	132
Frequency		Hz	50
Number of Phases			3
Neutral Earthing			Solid
Circuit Breaker Characteristics			
Number of poles			3
Class – indoor/outdoor	1.2		Outdoor
Rated voltage	4.1	kV	145
Rated Insulation level. Lightning impulse level	4.2	kV	550 630 (across isolating gap)
Power frequency withstand			230 265 (across isolator gap)
Rated Frequency	4.3	Hz	50
Rated Normal Current	4.4	A	2000
Rated Short-time withstand current	4.5	kA	40
Rated peak withstand current	4.6	kA	100
Rated duration of short circuit	4.7	S	3
Rated supply voltage of opening and closing devices and auxiliary and control circuits. Closing and tripping Indication Control	4.8	V	110 110 110
Rated supply frequency of closing and auxiliary circuits.	4.9	Hz	DC
Rated short-circuit breaking current. Equal to rated short-time withstand current.	(Part3)4.101	kA	40 @ X/R of 14
Rated short-circuit making current	(Part3) 4.103	kA	Manufacturer to state
Rated operating sequence.	(Part3)4.104		0 – 0.3 s – CO – 3min – CO
Characteristics for short line faults	(Part3)4.105		Manufacturer to state
Out of phase making and breaking.	(Part3)4.106		Manufacturer to state

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Rated capacitive switching currents : Rated cable-charging current.	(Part3)4.107		Manufacturer to state but should be Class 2
Rated small inductive breaking current.	(Part3)4.108		Manufacturer to state
Rated DC time constant.	(Part3)4.109		Manufacturer to state (see specified fault breaking current)
Classification of mechanical operations M1 or M2	(Part3)4.110		M1
Forced opening: Yes/No	(Part3)5.4.103		Manufacturer to state
Anti pumping: Yes/No	(Part3)5.5.101		Manufacturer to state
Mechanism type			Manufacturer to state
Circuit breakers with independent drive mechanisms	(Part3)5.5.103		Manufacturer to state
Local manual operation: Yes/No	(Part3)5.8.201		Manufacturer to state
Slow closing operation	(Part3)5.8.202		Manufacturer to state
Noise	(Part3)5.203		Manufacturer to state
External bushing creepage distance	5.14		As per 41-37 Pt 1 (25mm per kV)
External bushing pollution performance	5.14.101		As per 41-37 Pt 1 (Class III)
Personal safety clearance	5.107		As per 41-37 Pt 1 (i.e. additional 300mm)
Air insulated bushings	5.108		As per 41-37 Pt 1
Provision of key interlock			Provision for 1 key
CT secondary terminal			Spring loaded or "cage clamp" design. Spring washers on stud type
Bus zone CT secondary terminals.			Shorting/disconnecting facility.
Scada terminals			Knife terminal equipped with test 2.3mm sockets

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APPENDIX 2: Declaration of Technical Performance to be completed by the manufacturer

	MANUFACTURER			
	CIRCUIT BREAKER TYPE REFERENCE			
		ENATS 41-37 Reference	UNIT	VALUE
	CIRCUIT BREAKER CHARACTERISTICS			
1	Number of poles			
2	Class – indoor/outdoor	1.2		
3	Rated voltage	4.2	kV	
4	Rated Insulation levels. Lightning impulse: Nominal Across isolating gap Power frequency: Nominal Across isolating gap		kV	
5	Rated Frequency	4.3	Hz	
6	Rated Normal Current	4.4	A	
7	Rated Short-time withstand current (3 second)	4.5	kA	
8	Rated peak withstand current	4.6	kA	
9	Rated duration of short circuit	4.7	S	
10	Rated short-circuit breaking current.	(Part3)4.101	kA	
11	Rated short-circuit making current	(Part3) 4.103	kA	
12	Rated operating sequence.	(Part3) 4.104		
13	Characteristics for short line faults	(Part3) 4.105		
14	Out of phase making and breaking.	(Part3) 4.106	kA	
15	Rated capacitive switching currents : Rated line charging breaking current. Rated cable-charging breaking current. Rated Single Capacitor bank breaking current. Class.	(Part3) 4.107	A A A	
16	Rated small inductive breaking current.	(Part3) 4.108	A	
17	Rated DC time constant.	(Part3) 4.109	Msec	
18	Classification of mechanical operations	(Part3) 4.110	M1/M2	
19	Electrical endurance	(Part3) 4.111	E1/E2	

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20	Number of breaks/phase		-	
21	Type reference of vacuum-bottle or interrupter, as applicable.		-	
22	Maximum main circuit resistance.		Ohms	
23	Mass of complete unit		Kg	
24	Noise	(Part3)5.203	dB	
	OPERATING MECHANISM			
25	Manufacturer and Type Reference of Operating Mechanism.		-	
26	Mechanism travel.		Mm	
27	Forced opening	(Part3)5.4.103	Yes/No	
28	Anti pumping	(Part3)5.5.101	Yes/No	
29	Local manual operation	(Part3)5.8.201	Yes/No	
30	Slow close/open safety interlocking	(Part3)5.8.202	Yes/No	
31	Rated supply voltage of opening and closing devices and auxiliary and control circuits: Closing and tripping Indication Control		V	
32	External bushing creepage distance	5.14	mm	
33	External bushing pollution performance	5.14.101	Class	
34	Personal safety clearance: Height of HV connection above ground.	5.107	mm	
35	Rated supply frequency of closing and auxiliary circuits.		Hz	
36	Closing coils: Number and type Type reference Power consumption		W	
37	Tripping Coils: Number and type Type reference Power consumption		W	
38	Operating time – Close		Ms	
39	Operating time – Open		Ms	

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40	Key interlock		Yes/No	
41	Secondary wiring terminal connections: Control wiring terminals CT secondary wiring terminals Bus zone CT secondary terminals SCADA terminals		Manufac -turer and type ref	
42	Maximum dynamic floor loading		kN	
	SF6 GAS SYSTEM			
43	Design pressure of gas system at 20°C		Barg / MPa	
44	Operating pressure of pressure relief device		Barg / MPa	
45	Normal pressure at 20°C		Barg / MPa	
46	Alarm pressure		Barg / MPa	
47	Close lockout pressure		Barg / Mpa	
48	Trip (open) lockout pressure		Barg / MPa	
49	Minimum close pressure		Barg / MPa	
50	Minimum trip pressure		Barg / MPa	
51	Quantity of gas (mass and volume)		kg & litres	
52	Method of monitoring pressure and temperature compensation			
53	Moisture content at commissioning (max.) Dew point temperature at commissioning (max.)		mg/kg °C	
54	Maximum permissible acidity level (HF)			
55	Maximum permissible leak rate (<1% per annum)		% and litres	
56	Mass of alumina/molecular sieve and type reference.		kg	

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APPENDIX-3: Addendum to Supplier Requirements

Project specific installation and protection requirements will be provided by Primary Engineering Projects for inclusion in this appendix.